



THEORETICAL COMPUTER SCIENCE

SPECIAL ISSUE

Algebraic and Numerical Algorithms

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Symbolic/algebraic and numerical algorithms are the backbone of the modern computations in sciences, engineering and electrical engineering. These two classes of algorithms and the respective scientific communities have been separated historically and still have relatively little interaction; yet they may benefit from combining their power and techniques toward common computational goals. Such a combination is a rather recent undertaking, which became visible only in the last decade, but it is already popular in such central areas as root-finding for polynomials and systems of polynomials and computations with Toeplitz and other structured matrices (having ties and impact to both numerical matrix methods and algebraic polynomial techniques). The areas happened to be central also for the Editors' research interests, which motivated our efforts to bring the subjects to the attention of the TCS readers. (The readers can find links to the introductory and advanced bibliography on these subjects in the Editor's web home pages.)

The present issue of the TCS partly reflects the state of the art. It includes papers on various algebraic and numerical algorithms and techniques, focuses on combined application of the methods from both groups, and extensively represents topics in polynomial root-finding and structured matrix computations.

The issue includes recent advances in the numerical study of the root variety for a system of multivariate polynomials by homotopy methods, leading to the variety's irreducible decomposition and the approximation of all common roots, by Sommese et al. A complexity study is undertaken by Bompadre et al, for the problem of solving polynomial systems by combining different techniques such as the Newton-Hensel lemma, characteristic polynomials, and resultants, in the setting of Straight-Line programs. The same setting is used by Pardo and San Martín for the purpose of solving Pham systems by means of deformation techniques. The Issue also includes contributions on analysis of the condition number of sparse systems, based on symplectic geometry tools, by Malajovich and Rojas.

In the area of structured matrices, this Issue includes novel algebraic Toeplitz-like matrix structure found and exploited by Bini and Gemignani to accelerate numerical computation of polynomial greatest common divisors (stabilized by the transition to Bernstein's basis); numerical inversion of triangular Toeplitz matrices, equivalent to computing polynomial reciprocals, by Lin et al; algebraic techniques for handling degeneracies of skew-symmetric Toeplitz matrices by Heinig and Rost; algebraic methods for approximate solution of the inverse eigenproblem for structured matrices (motivated by the design of neural networks) by Bai and Chan; novel techniques of least-squares compression of structured matrices applied in Newton's iteration for their inversion by Pan et al; and the study of the matrix algebras of the preconditioners for multilevel Toeplitz systems by Noutsos et al.

Kaporin re-examines the known algebraic algorithms for fast matrix multiplications and modifies them for practical numerical computing, with quite surprising and promising results.

The *abc* conjecture Number Theory is employed by Croot et al, in order to reduce the number of bits required in rounding the reciprocal square root.

Topics in complexity analysis and geodesic constructions from a computational geometry point of view are considered by Burago et al, and approximation algorithms are proposed.

An important application of parameterizing algebraic curves is treated by Pérez-Díaz et al, where a numerical singularity is exploited to achieve a good parameterization by lines.

The Issue includes a fundamental work by Gathen and Nöcker on the acceleration of the computations in finite fields with normal bases, by using fast polynomial arithmetic.

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